

Prompt 1 — Load + Inspect

Write a Python script that loads a CSV file called "data.csv".

Print:

- First 5 rows
- Column names
- Data types

Handle file not found errors gracefully.

Keep code simple and readable.

Prompt 2 — Cleaning

Extend the script to clean the dataset:

- Drop rows with missing values in:
county, analyte_name, maximum_concentration, sdwa_limit, year
- Convert:
maximum_concentration → float
sdwa_limit → float
year → integer
- Print:
 - Number of rows before cleaning
 - Number of rows after cleaning

Prompt 3 — Filter (Arsenic Only)

Filter dataset:

- Keep only rows where analyte_name == "Arsenic"

Print:

- Number of rows after filtering
- Unique counties remaining

Prompt 4 — Aggregation

For each county:

- Find MOST RECENT year available
- Filter to that year

Compute:

- max_concentration (worst-case)
- avg_concentration
- sdwa_limit

Store in dictionary:

```
{  
  county: {  
    "max_concentration": value,  
    "avg_concentration": value,  
    "year": value,  
    "sdwa_limit": value  
  }
```

```
}  
}
```

Print one example.

Prompt 5 — Risk Calculation

Add risk calculation:

```
risk_score = min(100, (max_concentration / sdwa_limit) * 100)
```

Assign category:

- 0-40 → Well Below Limit
- 40-75 → Moderate
- 75-100 → Approaching Limit
- 100 → Above Limit

Add to results.

Prompt 6 — Confidence

Add confidence score:

- ≤1 year → High
- 1-3 years → Medium
- >3 years → Low

Add to results.

Prompt 7 — Fallback Handling

Add fallback logic:

```
If max_concentration missing:  
    use avg_concentration  
    fallback_used = True  
Else:  
    fallback_used = False
```

Add fallback_used field.

Prompt 8 — Explanation

Generate a plain English explanation:

- Start with "Is my water safe?"
- Include "This does NOT necessarily mean unsafe water"
- Avoid jargon
- Keep to 2-3 sentences

Prompt 9 — Guidance

Add a guidance field:

Well Below → No action needed
Moderate → Monitor
Approaching → Consider filtration
Above → Consider testing

Keep neutral tone.

Prompt 10 — Final Output

Format results as list of dictionaries with:

county, risk_score, category, max_concentration,
avg_concentration, explanation, guidance,
confidence, fallback_used

Prompt 11 — Export JSON

Save results to "output.json" with indent=2.

Prompt 12 — UI Architecture

Build extensible UI:

- County dropdown
- Contaminant dropdown (Arsenic active, others disabled)
- Clean layout
- One HTML file

Prompt 13 — Core UI Display

Display:

- Risk score (large)
- Category
- Explanation
- Guidance

Focus on clarity and interpretation

Prompt 14 — Data Transparency

Show:

- Max (worst-case)
- Avg (typical)
- EPA limit

Prompt 15 — Trust Layer

Show:

- Confidence
- Year
- Limitation note
- Fallback message if used

Prompt 16 — Multi-Contaminant Design

Make UI reusable:

- Generic labels
- Dynamic contaminant
- Placeholder for unsupported contaminants

Prompt 17 — UI Polish

Improve:

- Typography
- Colors (green/yellow/orange/red)
- Spacing
- Readability <5 seconds